Endo-HJEPA: A Hierarchical Joint-Embedding World Model for Unified Endoscopic Video

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Abstract

Endoscopic artificial intelligence is predominantly reactive, whereas collision warning and assisted navigation require anticipating how the scene changes under candidate actions. Pixel prediction is poorly matched to endoscopy because highlights, fluid and smoke dominate an intrinsically uncertain image residual. We introduce Endo-HJEPA, a hierarchical joint-embedding world model that predicts representations rather than pixels across laparoscopy, gastrointestinal endoscopy and bronchoscopy. A frozen V-JEPA 2 ViT-L encoder is coupled to a causal residual forecaster, a temporally coarse predictor, an action-conditioned predictor over vector-quantised latent changes, and a contrastive transition-energy head used for sampling-based latent model-predictive control. On a video-level evaluation drawn from 1,707 sequences and 19 datasets, the causal forecaster reaches 0.978 cosine at horizon four, compared with 0.916 for persistence, 0.974 for a GRU and 0.971 for Mamba (Holm-corrected Wilcoxon $p < 10^{-80}$). Performance improves from 500 to 6,000 training clips and then plateaus at 13,552 clips. Latent planning beats persistence on 98.0% of held-out clips, while transition energy correlates with SCARED camera-to-tissue proximity (AUC 0.682). However, navigation to arbitrary physical targets succeeds in only 10% of trials, and unsupervised latent actions are not semantically grounded. These results support hierarchical representation prediction as a reproducible basis for endoscopic forecasting, but not yet as an autonomous controller. The evaluation code, split protocol and negative results are released.

Keywords: world model, joint-embedding predictive architecture, endoscopy, representation learning, latent planning, self-supervised learning

1. Introduction

1.1. Clinical context: endoscopy needs anticipation, not only recognition

Minimally invasive diagnosis and therapy are increasingly delivered through an endoscope—laparoscopes in the abdomen, flexible scopes in the gastrointestinal tract, and bronchoscopes in the airway. Across all three orifices, safe operation depends on a clinician's ability to *anticipate*: how tissue will deform under contact, how a lumen will open as the scope advances,

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where an instrument will move, and when the camera is about to collide with a wall or lose the view. Current endoscopic AI is almost entirely *reactive*. It recognises the present frame—surgical phase, instrument presence, anatomy—but cannot forecast what comes next, nor what a candidate camera adjustment would produce. A model that predicts how an endoscopic scene will evolve, and how it would evolve under a candidate action, is a prerequisite for anticipatory guidance, collision warning, and any future assisted scope control. That is the clinical target of this paper: not a stronger phase classifier, but a *world model* that can be queried about the future.

1.2. The technical problem: structure is predictable, appearance is not

A world model for endoscopy faces a specific asymmetry. The *structure* of endoscopic video—lumen geometry, camera advance, instrument trajectories—follows smooth physical dynamics and is strongly predictable. The *appearance*—specular highlights, fluid, smoke, mucosal micro-texture, peristalsis—is high-entropy and largely unpredictable. The dominant world-model paradigm, pixel generation (diffusion world-action models, Genie-style token video), must spend capacity forecasting this unpredictable appearance. That both wastes capacity and ties the model to a pixel metric that is dominated by glare and fluid rather than by the quantities that matter for planning. Joint-embedding predictive architectures (JEPA) offer the alternative we adopt: predict in a learned representation space, where unpredictable appearance residual can be ignored by design, and where the predicted quantities are exactly the plannable ones.

Key insight: predict plannable representations, not the next frame.. If a frame is written as structure plus appearance, any pixel predictor is lower-bounded by the appearance dispersion, regardless of how well structure is modelled (Section 3.8). A representation that is sufficient for structure and invariant to appearance has no such bound. Endoscopy is an unusually clean instance of this split: the camera moves through a deformable lumen under clinician control (predictable), while highlights and fluid dominate the pixel residual (unpredictable). The design consequence is that a world model should (i) forecast in encoder space, (ii) separate fast dynamics from slow anatomy, (iii) score which futures are plausible, and (iv) plan by rolling those futures—without ever decoding pixels.

1.3. Gap

Despite rapid progress in surgical video understanding, two gaps remain. **(i) Single scale.** Existing surgical JEPA and video-pretraining systems (SurgRec-JEPA, JHU-VPT, VideoMAE) learn a single temporal predictor and do not separate fast tissue/tool/camera dynamics from slow anatomical and procedural structure. LeCun's hierarchical JEPA (H-JEPA) argues such separation is essential for long-horizon prediction and planning, but it has not been instantiated for endoscopy. **(ii) Per-orifice fragmentation.** Endoscopic models are built per orifice—laparoscopy, GI endoscopy, and bronchoscopy are treated as separate foundation-model problems—even though all three share the same underlying physics of a camera advancing through a deformable lumen under clinician control. A single domain-conditioned world model is both a stronger scientific claim and a more data-efficient one, but it requires evidence that such sharing is beneficial, or necessary.

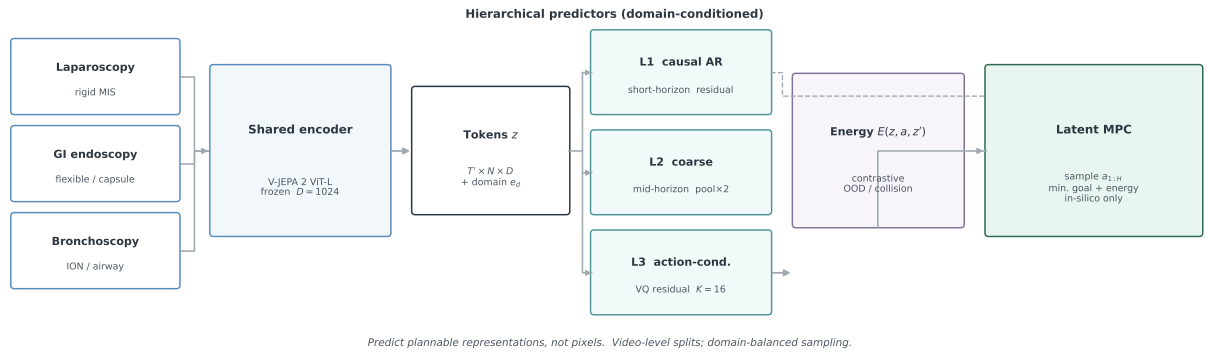

Figure 1: Overview of Endo-HJEPA. Three orifice streams share a frozen V-JEPA 2 ViT-L encoder. Domain-conditioned L1/L2/L3 predictors forecast future tokens at increasing temporal abstraction. A contrastive energy head scores transition plausibility and drives latent MPC. Boxes hold only the module name and the governing variable; no pixels are decoded. When a goal latent is unavailable, the system reduces to L1 forecast plus energy-based rejection.

1.4. Our approach

We instantiate this idea in **Endo-HJEPA** (Figure 1). A shared frozen V-JEPA 2 ViT-L encoder maps clips from all three orifices to dense tokens. Three domain-conditioned predictors operate at increasing temporal abstraction: L1 (causal autoregressive, residual, short horizon), L2 (temporally pooled, mid horizon), and L3 (action-conditioned over a residual codebook). A contrastive energy head scores (z, a, z') transitions and is used both as an out-of-distribution prior and as the cost in sampling-based latent MPC. The encoder is frozen by default so that every dynamics ablation shares identical latents; an optional last-block adaptation is evaluated only on recognition probes. No decoder is trained. The evidence is scoped as follows: forecast and scale are measured on a 6,000-clip causal-L1 protocol; planning, energy, and physical probes use the full H-JEPA trained at 2,000 clips; recognition uses frozen versus adapted V-JEPA 2 on the official CholecT50 split. These protocols are never mixed in a single table.

1.5. Contributions and significance

1. **A hierarchical, energy-based JEPA world model for unified cross-orifice endoscopy.** We instantiate H-JEPA (short-horizon residual dynamics, mid-horizon anatomy, action-conditioned planning with an energy prior) on the official V-JEPA 2 ViT-L encoder, trained and evaluated across laparoscopy, GI endoscopy, and bronchoscopy with one domain-conditioned dynamics model.
2. **A causal-autoregressive latent forecaster.** The *form* of the latent forecaster matters more than its size: a GPT-style causal Transformer surpasses a strong GRU and a Mamba/SSM baseline, whereas a parallel query-token Transformer of the same capacity does not.
3. **A physically correlated uncertainty signal.** The contrastive energy head flags out-of-distribution transitions and correlates with camera-to-tissue distance on SCARED (near-wall AUC 0.682), not only with latent self-consistency.

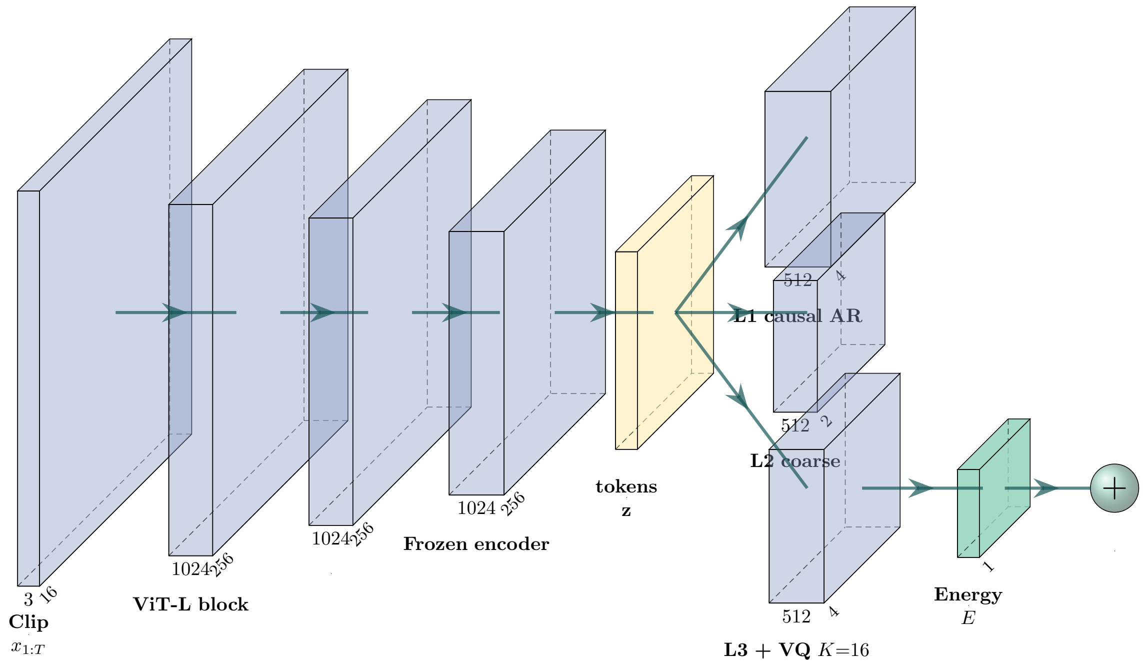

Figure 2: PlotNeuralNet rendering of the same stack: input clip, frozen ViT-L blocks, token tensor, L1/L2/L3 predictors, energy head, and latent-MPC aggregator. Colours are muted (print-safe blue/teal); block sizes encode relative tensor rank, not a CNN.

4. **A leakage-free, per-dataset protocol.** Video-level splits, domain-balanced sampling, per-dataset forecast decomposition, reproducible external baselines, and paired Wilcoxon tests with Holm correction, released as executable code.

We also report two characterised negative results: self-supervised latent actions are not semantically or physically grounded without encoder-level supervision, and zero-shot cross-orifice transfer fails but is recoverable with a 32-shot domain token. Together these provide evidence that unified multi-orifice training is necessary, not merely convenient. We do *not* claim the first surgical world model, pixel-level video generation, interpretable latent actions, uniform recognition SOTA, or any clinical efficacy endpoint.

Significance.. Endo-HJEPA reframes endoscopic AI from reactive recognition to anticipatory world modelling. Three properties matter clinically: anticipation (forecast is the primitive behind collision warning), a grounded uncertainty signal (energy tracks physical proximity), and unification (one domain-conditioned model serves three orifices where labelled video is scarce). All planning results are in-silico.

2. Related work

We position Endo-HJEPA along five axes: the world-model lineage it builds on, the surgical/endoscopic representation learning it extends, the surgical world models it differs from, the general video/image backbones we benchmark against, and the endoscopic datasets that anchor our evaluation.

2.1. World models and joint-embedding prediction

The idea of a learned world model for prediction and planning originates with Ha and Schmidhuber's World Models [1] and the PlaNet latent-dynamics planner [2], and was scaled to diverse control domains by Dreamer [3]. These models predict a learned latent state and plan by rolling it forward, but they are trained with a reconstruction or reward signal and target control, not video understanding. Energy-based formulations [4] provide the formal basis for scoring state compatibility without a generative decoder. The joint-embedding predictive architecture (JEPA) of LeCun's position paper [5] argues that predicting in representation space—rather than pixels—is the key to capturing predictable structure while ignoring unpredictable appearance; I-JEPA [6] realised this for images, V-JEPA [7] for video, and V-JEPA 2 [8] added scale and the action-conditioned (V-JEPA 2-AC) planning variant. Generative interactive environments such as Genie [9] take the opposite, pixel-generative route. A common limitation of these works, for our purpose, is that they are either single-scale (no temporal hierarchy), pixel-generative (capacity spent on unpredictable appearance), or evaluated on generic/control video rather than endoscopic dynamics. Endo-HJEPA adopts the JEPA principle and adds the hierarchy and energy prior that the surgical setting lacks.

2.2. Self-supervised surgical and endoscopic video representation

Self-supervised pretraining has become the default for surgical video. SurgRec-JEPA [10] and JHU-VPT [11] apply JEPA-style and video pretraining to surgical data and report strong downstream transfer, and endoscopic foundation models such as Endo-FM [12] scale self-supervised pretraining to large endoscopic corpora. These methods produce a single-scale representation for *recognition* (phase, tools, segmentation); none of them build a dynamics model that can forecast or plan. Endo-HJEPA shares their self-supervised, frozen-encoder efficiency but adds the temporal hierarchy and the action-conditioned planning layer that recognition-oriented SSL omits.

2.3. Surgical and endoscopic world models

World models tailored to surgery are emerging. The Surgical Vision World Model [13] generates pixel-controllable surgical video; SurgWorld [14] learns latent actions for cataract surgery; EndoWAM [15] applies diffusion world-action modelling to endoscopic navigation; and Surgical WAM [16] couples generative video with robot action chunks for closed-loop control. These are predominantly *pixel-generative* or *single-task*; they neither separate temporal scales nor provide an explicit uncertainty/energy signal for safe planning. Our contribution is orthogonal and complementary: a hierarchical JEPA with energy-driven latent MPC that predicts representations, not pixels.

2.4. General video and image representation backbones

For the representation evaluation we benchmark against the strongest *reproducible* general-purpose encoders. For video: VideoMAE [17] (Kinetics masked autoencoding), TimeSformer [18] (divided space-time attention), and ViViT [19] (factorised space-time); state-space video models such as VideoMamba [20] and the underlying Mamba block [21] are the recurrent/SSM reference for dynamics. For images: the ImageNet-supervised ViT [22] and DINOv2 self-supervised features [23], pooled over time. None of these provide a world-model predictor or a planning/energy capability, which is the gap Endo-HJEPA fills; §4 measures them on the same CholecT50 protocol.

Table 1: Principal notation (Section 3).

Symbol	Meaning
$x_{1:T}$, d	Input clip of T frames; orifice $d \in \{\text{laparo}, \text{gi}, \text{bronch}\}$
f_θ, $z_{1:T'}$	Frozen encoder; dense tokens $z \in \mathbb{R}^{T' \times N \times D}$
T', N, D	Tubelet steps ($T/2$), spatial tokens (256), width (1024)
e_d	Learnable domain embedding
H, H_{hist}	Forecast horizon and history (default 4)
$a_t \in \{1, \dots, K\}$	Discrete latent action (K=16)
$r_t = z_{t+1} - z_t$	Residual used by VQ and residual prediction
$E(z, a, z')$	Scalar transition energy (low = plausible)
s_i	Learnable log-variance of loss term i
$z^\star$	Goal latent for MPC
w	Specular / instrument tubelet weights

2.5. Endoscopic datasets and physical supervision signals

Our evaluation spans the standard endoscopic corpora: Cholec80 (phase + tool presence) [24], CholecT50 (action triplets + phase) [25, 26], EndoVis 2017/2018 (instrument segmentation) [27], CholecSeg8k (semantic segmentation) [28], endoscapes (segmentation + critical view of safety) [29], the GI corpora HyperKvasir [30] and Kvasir-Capsule [31], and the pose/depth corpora SCARED [32] and C3VD [33], with STIR [34] for deformable point tracks. To our knowledge, using SCARED/C3VD camera poses and STIR point tracks as *physical anchors* for latent actions, energy, and deformation—rather than as reconstruction targets—is a novel use of these supervision signals within a JEPA world model.

Summary.. Across the five axes, the field offers strong single-scale SSL representations, pixel-generative surgical world models, and general video/image backbones—but no hierarchical, energy-based JEPA world model trained and evaluated as a unified cross-orifice endoscopic system with a physically grounded uncertainty signal. That is the gap Endo-HJEPA occupies.

3. Method

Core idea.. Endo-HJEPA is a world model that answers two queries without generating pixels: (i) given a history of endoscopic tokens, what is the likely future representation; (ii) given a candidate latent action, is the implied transition plausible, and does it approach a goal representation. The first query is L1/L2 forecast; the second is L3 plus energy plus latent MPC. Residual anchoring and uncertainty weighting belong to the core pooled model. Dense VICReg, appearance weighting and STIR supervision are explicitly separated below as optional spatial or encoder-adaptation extensions; they are not silently mixed into the pooled forecast numbers. Figure 1 is the data flow; Figure 2 is the tensor-level stack; Table 1 lists symbols.

3.1. Problem formulation and assumptions

A clip $x_{1:T}$ from orifice d is mapped by a frozen encoder f_θ to

$$z_{1:T'} = f_\theta(x_{1:T}) \in \mathbb{R}^{T' \times N \times D}, \qquad T' = T/2,\ N = 256,\ D = 1024. \tag{1}$$

The world model is a collection of predictors g_ϕ on these tokens. It never decodes pixels. We factor g_ϕ into three temporal scales, mirroring H-JEPA: L1 forecasts $\hat{z}_{t+1:t+H}$ (tissue, tool, camera); L2 forecasts a stride-2 pooled stream (anatomy / phase); L3 forecasts the consequence of a discrete action a_t. A scalar energy $E(z_t, a_t, z_{t+1})$ scores transition plausibility. Planning is sampling-based MPC:

$$\hat{a}_{1:H} = \arg\min_{a_{1:H}} \left\| \hat{z}_{t+H} - z^\star \right\|_2^2 + \sum_{k=0}^{H-1} E(\hat{z}_{t+k}, a_{t+k}, \hat{z}_{t+k+1}). \tag{2}$$

Assumptions. (i) Endoscopic dynamics are approximately Markovian in the learned latent space at the tubelet timescale. (ii) The frozen V-JEPA 2 encoder is more predictable for structure than for appearance (tested by the pixel-generation contrast in §4.2). (iii) Orifice domains share enough low-level dynamics that a domain-conditioned predictor can serve all three, while differing enough that naive zero-shot transfer fails (tested in §4.4).

3.2. Shared encoder and optional endoscopic token weighting

We use `facebook/vjepa2-vitl-fpc64-256` (ViT-L, patch 16, tubelet 2, input 256^2), frozen by default. An optional flag unfreezes the last K blocks for recognition-oriented domain adaptation; that path is never mixed into the forecast tables. High-luminance, low-saturation tubelets (glare, fluid highlights) are weakly predictable and are down-weighted to a floor $w_{\min}$=0.25. When EndoVis instrument masks are available, instrument tubelets are up-weighted by $1 + \beta\, m_{\text{inst}}$. This mask-weighted loss is an optional scratch-JEPA/encoder-adaptation path, used for the mask analysis rather than the official frozen-V-JEPA 2 pooled forecast protocol. The main 6,000-clip and 13,552-clip forecast models therefore use uniform latent weights. Making that distinction avoids attributing their gains to supervision that they did not receive.

3.3. Hierarchical predictors

All predictors are Transformer-based. A shared domain embedding e_d is added to the history so that one set of weights serves three orifices.

L1, causal autoregressive (default pooled forecaster).. A GPT-style causal Transformer predicts the next pooled token from the causal context and feeds the prediction back for H steps. Let $h_t = \text{in}(z_t) + e_d + p_t$ be the projected history with positional encoding p_t, and let M be the upper-triangular causal mask. One rollout step is

$$\hat{z}_{t+1} = z_t + \text{head}\big(\text{Transformer}(h_{\le t}; M)\big)_t, \tag{3}$$

i.e. the network predicts a *residual* from the last observed (or last predicted) token. This matches the near-Markov, smooth structure of endoscopic camera motion. Section 4.2 shows that this inductive bias, not parameter count, is what lets the Transformer surpass a GRU: a parallel query-token Transformer of the same size does not.

L1, dense spatio-temporal.. For spatially resolved prediction a factorised block applies temporal attention per spatial site, then spatial attention per tubelet step, over all N=256 tokens. Learned positional embeddings are added along both axes; H query steps are appended along time. This captures global camera motion that independent per-site prediction cannot represent. In the optional dense configuration, dense L1 is trained jointly with the pooled head so that its pooled evaluation path is not an untrained leftover. The principal scale and recurrent-baseline experiments use the pooled causal configuration; dense prediction is a spatial extension, not the source of the reported 0.978 pooled score.

Residual (delta) prediction.. Consecutive endoscopic latents are highly correlated. Every predictor therefore forecasts the change from the last observed token,

$$\hat{z}_{t+k} = z_{T'} + f_\phi(\cdot)_k, \tag{4}$$

rather than an absolute future. The model can only improve on persistence; without residual anchoring a 70M predictor is high-variance and can fall below persistence (Section 4.8).

L2, coarse mid-horizon.. L1 tokens are average-pooled along time with stride 2; a second Transformer predicts the coarser future. L2 adds little to short-horizon pooled cosine; its role is a seconds-scale handle for anatomy and phase.

L3, action-conditioned.. Residuals $r_t = z_{t+1} - z_t$ are vector-quantised into a codebook of K=16 entries (straight-through estimator, commitment loss). The L3 predictor conditions on the resulting discrete ids. These ids are a *planning device*: they index visual change, not surgical verbs (Section 4.5).

Energy head.. $E(z, a, z')$ is an MLP on $[z; \mathrm{emb}(a); z']$, trained contrastively with a rolled negative z^-:

$$\mathcal{L}_E = \mathrm{softplus}\big(E(z_t, a_t, z_{t+1}) - E(z_t, a_t, z^-)\big). \tag{5}$$

Low energy marks in-distribution, predictable futures. High energy is used as a reject / collision-risk prior.

Latent MPC.. We sample N_{mpc}=32 action sequences of length H, roll L3, and select the sequence minimising (2). This is performed entirely in latent space and is in-silico only.

3.4. Objective

Each active core loss $\mathcal{L}_i$ carries a learnable log-variance s_i (uncertainty weighting), contributing $e^{-s_i}\mathcal{L}_i + s_i$. This prevents a hierarchy level from dominating only because of its numerical scale. The pooled full-model objective is

$$\mathcal{L}_{\mathrm{pool}} = \sum_{i \in \{\mathrm{L1,L2,L3},E,\mathrm{commit}\}} \big(e^{-s_i}\mathcal{L}_i + s_i\big). \tag{6}$$

The optional dense configuration adds a dense L1 term, a separately weighted pooled-L1 term and a VICReg-style anti-collapse regulariser on $\hat{z}$,

$$\mathcal{L}_{\mathrm{vr}} = \frac{1}{D}\sum_j \max\big(0,\, 1 - \mathrm{std}(\hat{z}_{:,:,j})\big) + \frac{1}{D}\sum_{i \neq j} \mathrm{Cov}(\hat{z})_{ij}^2. \tag{7}$$

Algorithm 1 Endo-HJEPA training and latent MPC (no pixel loss).

Training: for each domain-balanced clip $x_{1:T}$ with domain d:
  $z_{1:T'} \leftarrow f_\theta(x_{1:T})$
  $\mathcal{L}_1 \leftarrow \mathrm{SmoothL1}(\text{causal-L1}(z_{\mathrm{hist}}, e_d), z_{\mathrm{fut}})$ (residual)
  $\mathcal{L}_2 \leftarrow \mathrm{SmoothL1}(\mathrm{L2}(\mathrm{pool}(z_{\mathrm{hist}}), e_d), \mathrm{pool}(z_{\mathrm{fut}})[::2])$
  $\mathrm{ids}, \mathrm{commit} \leftarrow \mathrm{VQ}(z_{t+1} - z_t)$; $\mathcal{L}_3 \leftarrow \mathrm{SmoothL1}(\mathrm{L3}(z_{\mathrm{hist}}, \mathrm{ids}, e_d), z_{\mathrm{fut}})$
  $\mathcal{L}_E \leftarrow$ contrastive energy (5)
  $\mathcal{L} \leftarrow \mathcal{L}_{\mathrm{pool}}$; AdamW on ϕ
Planning (in-silico): given z_{hist}, goal $z^\star$, domain d:
  sample N_{mpc} action sequences; roll L3; return $\arg\min$ of (2)

so that $\mathcal{L}_{\mathrm{dense}} = \mathcal{L}_{\mathrm{pool}} + e^{-s_{\mathrm{L1d}}}\mathcal{L}_{\mathrm{L1d}} + s_{\mathrm{L1d}} + \lambda_{\mathrm{vr}}\mathcal{L}_{\mathrm{vr}}$. STIR fine-tuning is evaluated as a separate encoder-adaptation experiment rather than presented as part of the core pooled checkpoint. Persistence (copy last latent) is a mandatory baseline for every forecast table.

3.5. STIR point-track regulariser

STIR provides IR start/end segmentations rather than dense tracks. We sample up to 64 points from each mask, map them onto the token grid, and apply a symmetric chamfer between tokens at those sites at the first and last timestep:

$$\mathcal{L}_{\mathrm{STIR}} = \mathrm{Chamfer}\big(z_0[p_{\mathrm{start}}],\ z_{T'-1}[p_{\mathrm{end}}]\big). \tag{8}$$

This optional loss adapts the unfrozen encoder in representation space, with no pixel regression. The reported before/after STIR result comes from this separate fine-tuning run; it is not evidence that the main full-H-JEPA checkpoint was jointly trained with STIR.

3.6. Physical alignment of latent actions

SCARED stores a 4×4 camera pose per frame; C3VD provides `pose.txt`. The relative pose $T_t^{-1}T_{t+1}$ is mapped to a 6D twist. We quantify grounding by (i) normalised mutual information between k-means clusters of latent residuals and of twists, against a random-id chance baseline, and (ii) a closed-form linear probe from residual to twist (MAE, R^2). This is a *probe*, not a training loss, unless encoder-level verb supervision is explicitly enabled.

3.7. Training procedure

Clips are enumerated from a sequence-level manifest with video-level splits (hash of `dataset::sequence_id`); no clip crosses a split boundary. Sampling is domain-balanced by round-robin so laparoscopy does not dominate GI/bronchoscopy. Dense tokens are cached once per encoder; L1, full H-JEPA, GRU, and Mamba train on the *same* cached latents. Optimisation: AdamW (lr 3×10^{-4}, weight decay 0.01), cosine schedule with 5% warmup, gradient clip 1.0. Full H-JEPA: 69.7M trainable parameters; GRU: 2.63M; Mamba: 1.58M.

3.8. *Why representations, not pixels*

Write a frame as $x = (s, n)$: structure s (lumen, pose, instruments) with $s_{t+1} = g(s_t) + \varepsilon$ and $\mathbb{E}\|\varepsilon\|_2^2 = \sigma_s^2$ small; appearance n (highlights, fluid) drawn independently of the past with dispersion σ_n^2. A decoder $r(s, n)$ is Lipschitz in n with constant L_n.

Proposition 1 (pixel prediction is appearance-limited). Any next-frame predictor $\hat{x}_{t+1} = \hat{r}(x_{1:t})$ satisfies $\mathbb{E}\|x_{t+1} - \hat{x}_{t+1}\|_2^2 \geq \frac{1}{2} L_n^2 \sigma_n^2$: the Bayes predictor of independent appearance is its mean, so the pixel error is lower-bounded by appearance dispersion regardless of how well g is modelled.

Proposition 2 (representation prediction is structure-limited). If E is sufficient for structure and invariant to appearance, $E(s, n) = E(s)$, then a JEPA predictor on $z = E(x)$ has error $O(\sigma_s^2)$ with *no* dependence on σ_n^2.

These statements formalise the design choice; they do not by themselves prove downstream superiority. That is tested in §4.2, where a next-frame CNN (PSNR 17.9 dB) and a conditional DDPM (4.9 dB) both lose to copy-last (29.6–30.4 dB).

4. Experiments

Goal.. The experiments test the two queries of Section 3: can Endo-HJEPA forecast plannable endoscopic representations better than persistence and strong recurrent/SSM baselines, and can L3+energy+MPC plan and reject in a way that is physically meaningful. They are organised by six research questions that map onto the contributions, not by dataset dump. Unless marked "lit.", every number is measured from a named JSON under `outputs/` and copied into `docs/endohjepa/verified_metrics.json`; literature rows are never mixed into a measured comparison. Forecast tables use the 6,000-clip causal-L1 protocol; planning / energy / pose tables use the 2,000-clip full H-JEPA; recognition uses the official CholecT50 split. These protocols are labelled in every caption.

Organization.. Section 4.1 states data, metrics, baselines, and implementation. Section 4.2–4.2 report forecast and scale (C1). Section 4.3 reports planning and energy (C2). Section 4.4 reports cross-orifice transfer (C3). Section 4.5 reports action grounding (negative). Section 4.6 reports recognition probes (encoder quality, not the headline claim). Section 4.7 is the per-dataset decomposition. Section 4.8 isolates design choices.

4.1. *Datasets, metrics, baselines, implementation*

Datasets.. Table 2 lists all 19 local corpora (1,707 sequences, 1,067,734 decoded frames). Splits are video-level (hash of `dataset::sequence_id`): train 1,328 / val 188 / test 191 sequences. TrackVes and SurgT include video-only rows whose decoded-frame count is unknown; the census records those rows as zero rather than subtracting sentinel values, and they are excluded from latent forecast. Full Cholec80 is not local (CAMMA request; only videos 41–45 plus CholecT50). C3VD is restricted to `cecum_t1_a` by Drive quota. ION bronchoscopy is private and stored only as anonymised `case_XXX`.

Table 2: Dataset census (video-level split). Forecast uses decoded RGB clips; pose/depth/mask columns mark physical probes, not the forecast loss.

Domain	Dataset	Seq.	Frames	Role
GI	HyperKvasir [30]	768	96,567	forecast
GI	Kvasir-Capsule [31]	234	317,726	forecast
GI	Kvasir-Instrument	1	590	forecast
GI	C3VD [33]	1	1,379	pose probe
Laparo	Stereo_Lap	44	185,344	forecast
Laparo	endoscapes [29]	11	153,899	forecast
Laparo	CholecT50 [25]	50	100,863	forecast + phase/tools
Laparo	Cholec80-Boxes [24]	5	15,691	forecast (v41–45)
Laparo	CholecSeg8k [28]	1	8,080	semantic (not main)
Laparo	EndoVis 2017/2018 [27]	13	5,701	forecast + masks
Laparo	SCARED [32]	76	1,878	pose + depth
Laparo	STIR [34]	360	3,640	deformation
Laparo	tracking / own / EndoNeRF	67	12,933	forecast / tracking
Bronch	ION_bronch	76	163,443	forecast (private)
	Total	1,707	1,067,734	

Metrics.. Latent forecast is cosine similarity and MSE between predicted and target *pooled* tokens at horizon h,

$$\cos_h = \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} \frac{\langle \hat{z}^{(v)}_{t+h},\, z^{(v)}_{t+h} \rangle}{\|\hat{z}^{(v)}_{t+h}\|_2 \, \|z^{(v)}_{t+h}\|_2}, \tag{9}$$

averaged over video-level val clips $\mathcal{V}$. Higher cosine / lower MSE is better. Planning reach is the fraction of clips on which latent MPC is closer to the goal latent than persistence. Energy quality is the fraction of clips on which planned actions have lower energy than random actions, plus near-wall AUC on SCARED depth. Recognition uses video-level linear-probe phase accuracy and instrument mAP (mean $\pm$ std over 3 seeds). Action grounding uses NMI versus a random-id baseline.

Baselines.. Dynamics: persistence, GRU, Mamba/SSM, query-token L1. Representation: ImageNet ViT, DINOv2, VideoMAE, TimeSformer, ViViT, all on the same official CholecT50 split. Pixel contrast: next-frame CNN and conditional DDPM. V-JEPA 2-AC and SurgRec-JEPA have no public endoscopic-planning checkpoints; we cite their papers and do not invent a numeric comparison.

Statistics.. Paired bootstrap (1,000 resamples) and Wilcoxon signed-rank tests against GRU and Mamba, Holm-corrected across horizons, on 750 video-level clips.

Implementation.. Single GPU; encoder cached once; AdamW as in Section 3. Hyperparameters: hidden 512, 8 heads, 4 layers, dropout 0.1, VICReg weight 0.1, K=16, history/horizon 4/4. Varying depth (2–6), width (256–768) and lr (10^{-4}–10^{-3}) changes 6k-clip cosine by $<$0.002.

Table 3: Video-level latent forecast at h=4 (6,000-clip training, shared cache). Source: `outputs/scale_6000_*/val_metrics.json`.

Model	cos h=4	MSE h=4
Persistence	0.916	0.532
Endo-HJEPA (query-token L1)	0.936	0.408
Mamba / SSM dynamics	0.971	0.180
GRU dynamics	0.974	0.161
Endo-HJEPA (causal L1)	**0.978**	**0.139**

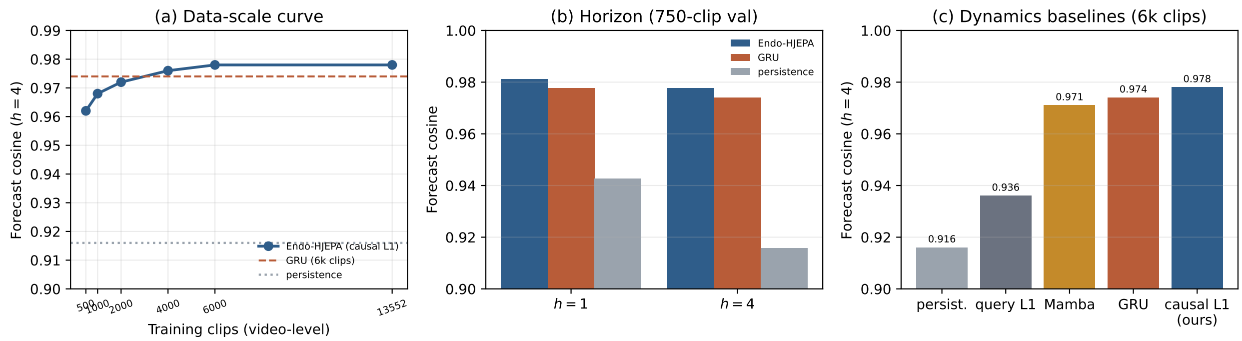

Figure 3: (a) Forecast cosine versus training clips; GRU and persistence are 6k-clip references. (b) Horizon h=1, 4 on the 750-clip val set used for Wilcoxon tests (Table 3 uses h=4 pooled cosine 0.978). (c) Same 6k-clip protocol as Table 3. Numbers are from `verified_metrics.json`.

4.2. RQ1–RQ2: Latent forecast and data scale

Table 3 and Figure 3 are the headline forecast result. Causal L1 exceeds persistence, query-token L1, Mamba, and GRU on the same 750-clip val set, with the lowest MSE. The margin over GRU is small in absolute cosine (0.004) but decisive in a paired test: at h=1, 0.9811 vs 0.9777 (Holm p=1.24×10^{-84}); at h=4, 0.9776 vs 0.9740 (p=4.06×10^{-102}). Against Mamba the h=4 Holm p is 1.23×10^{-111}. Predictor *form* is the lever: query-token L1 (70M) loses to GRU (2.6M); switching the same Transformer to causal autoregression plus residual (4) reverses the ranking.

The scale curve (Figure 3a) is monotonic from 500 to 6,000 clips (0.962 → 0.968 → 0.972 → 0.976 → 0.978) and stays above the 6k GRU throughout. Extending the same causal L1 to all 13,552 pooled clips that the local corpus yields leaves cosine flat at 0.978 and trims MSE from 0.139 to 0.135 (val n=1,631). The model is data-scalable up to a few thousand clips; beyond that, on this frozen encoder, forecast cosine has reached a plateau and further gains should come from method (dense spacetime L1, action supervision), not from stacking more of the same pooled tokens.

Pixel-generation baselines confirm Proposition 1: a next-frame CNN reaches 17.9 dB PSNR against copy-last 29.6 dB; a conditional DDPM reaches 4.9 dB against copy-last 30.4 dB (Figure 6a). Pixel error concentrates on unpredictable appearance; representation-space forecast does not have to.

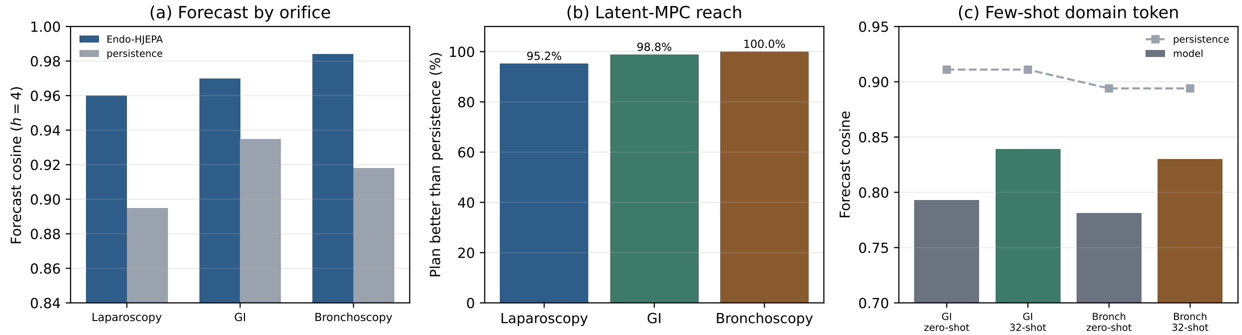

Figure 4: (a) Forecast by orifice on the 2,000-clip full model (same protocol as Table A.8). (b) Latent-MPC reach versus persistence. (c) Few-shot domain-token adaptation (32 target clips) from a laparoscopy-only model. Source: `p2000_full_causal/eval_ckpt_cumulative_energy.json`, `t16_transfer_laparo/fewshot_*.json`.

4.3. RQ3: Latent planning and physical energy

Planning uses L3+energy+MPC and has no GRU counterpart. On the 2,000-clip full model (250 val clips), latent MPC is closer to the goal latent than persistence on **98.0%** of clips (cosine 0.920 vs 0.874). By orifice: laparoscopy 95.2%, GI 98.8%, bronchoscopy 100% (Figure 4b). With the corrected cumulative rollout score in Eq. (2), planned trajectories have lower energy than independently sampled random trajectories on 92.8% of clips. On SCARED depth, energy flags near-wall transitions above chance (AUC 0.682, Spearman -0.471, n=132; Figure 6b). This is the physical reading of the energy prior: it correlates with camera-to-tissue distance, the quantity that matters for wall-contact safety.

Goal-directed navigation to an *arbitrary* SCARED viewpoint is a harder task: latent reach is 10% of 10 trials, with no pose-error reduction versus persistence. The world model predicts the *likely* future well; it cannot yet steer to an arbitrary anatomical target, because latent actions do not encode how to move (Section 4.5). We report this as a boundary, not as a success.

4.4. RQ4: Cross-orifice transfer

A laparoscopy-only model beats persistence in-domain (0.922 vs 0.850) but falls *below* persistence on zero-shot GI (0.791 vs 0.913) and bronchoscopy (0.776 vs 0.893). Fine-tuning only the domain embedding on 32 target clips recovers +0.045 (GI) and +0.049 (bronch) (Figure 4c). The unified model beats persistence in every domain (Table A.8). Orifice domains are distinct enough that zero-shot transfer fails; unified training, or a cheap domain token, is necessary.

4.5. RQ5: Action grounding (negative result)

Emergent latent actions are not physically grounded (SCARED pose NMI 0.41–0.52 versus random ≈0.40; residual→twist R^2 negative) and not semantically grounded (CholecT50 verb NMI 0.044 versus random 0.056; probe accuracy 0.306 versus majority-class 0.390). A codebook-level verb loss does not help. Encoder-level verb supervision lifts residual→verb NMI from chance 0.026 to 0.056—above chance, still weak. Latent actions index visual change that is useful for forecast and MPC; they are not discovered surgical actions.

Table 4: CholecT50 official *challenge-test* videos (VID68, 70, 73–75), video-level linear probe, mean $\pm$ std over 3 seeds. This is not the recommended five-fold cross-validation benchmark. Source: `outputs/vjepa2_adapted/cholect50_probe_multiseed.json`. We do not claim uniform dominance.

Encoder	Phase acc	Instrument mAP
ImageNet ViT (supervised)	0.592 ± 0.016	0.430 ± 0.003
DINOv2 (image SSL)	0.658 ± 0.016	$\mathbf{0.575 \pm 0.003}$
VideoMAE (Kinetics SSL)	0.558 ± 0.012	0.497 ± 0.004
TimeSformer (video ViT)	0.683 ± 0.012	0.490 ± 0.010
ViViT (video ViT)	0.692 ± 0.024	0.443 ± 0.004
V-JEPA 2 (frozen)	$\mathbf{0.704 \pm 0.006}$	0.406 ± 0.002
V-JEPA 2 (domain-adapted)	0.688 ± 0.057	0.485 ± 0.005

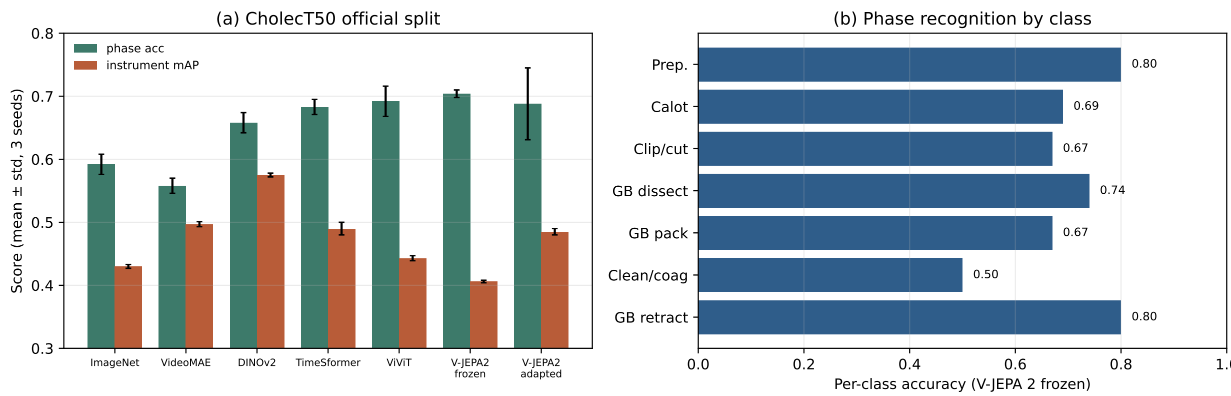

Figure 5: (a) Official challenge-test probes matching Table 4 (error bars = 3-seed std). (b) Per-class phase accuracy of frozen V-JEPA 2: cleaning/coagulation is hardest (0.50), preparation and retraction easiest (0.80).

4.6. RQ6: Downstream recognition

On the official challenge videos (VID68, 70, 73–75), frozen V-JEPA 2 is the strongest *video* model for phase (0.704), above ViViT, TimeSformer, DINOv2, ImageNet and VideoMAE. DINOv2 leads instrument presence (0.575); our adaptation improves instrument mAP over frozen V-JEPA 2 ($0.406 \to 0.485$) at a small, high-variance phase cost. This is an encoder-quality table, not a claim that the world model is a recognition SOTA. STIR fine-tuning reduces held-out deformation chamfer from 179.9 to 168.2 (-6.5%; Figure 6c).

Published SOTA context and non-comparability.. Table 5 states the comparison boundary explicitly. The matched claim of this paper is limited to the baselines rerun on our identical latent or challenge-test protocols. Published task-specific systems use different labels, supervision and splits, and are included to identify the remaining gap rather than to manufacture a ranking.

4.7. Per-dataset decomposition

"Per-domain" (three orifices) hides dataset-level variation. We reconstruct the same 750-clip video-level val list used to build the 6,000-clip cache (`domain_balanced_indices`,

Table 5: Position relative to published task-specific results. CV=cross-validation; SR=closed-loop success rate. Values in different protocol columns must not be ranked directly.

Task	Published reference	Endo-HJEPA evidence	Valid conclusion
CholecT50 instrument presence	RDV 0.894±0.020 mAP, official 5-fold CV [26]	0.485±0.005, five-video challenge test, linear probe	Not directly comparable; no SOTA claim
Cholec80 online phase	SurgicalMamba 94.6% accuracy [35]	Full Cholec80 is unavailable locally	Required external benchmark remains open
Physical endoscope navigation	EndoWAM 80.2% SR on 96 phantom trials [15]	10% reach on 10 offline SCARED trials	Demonstrates the present control gap
Surgical video generation	SurgWM reports PSNR/SSIM/FVD [13]	No decoder; latent cosine and energy	Orthogonal objectives and metrics
Latent endoscopic forecast	No standard public benchmark identified	0.978 cosine versus matched persistence/GRU/Mamba	Strong internal comparison, not external SOTA

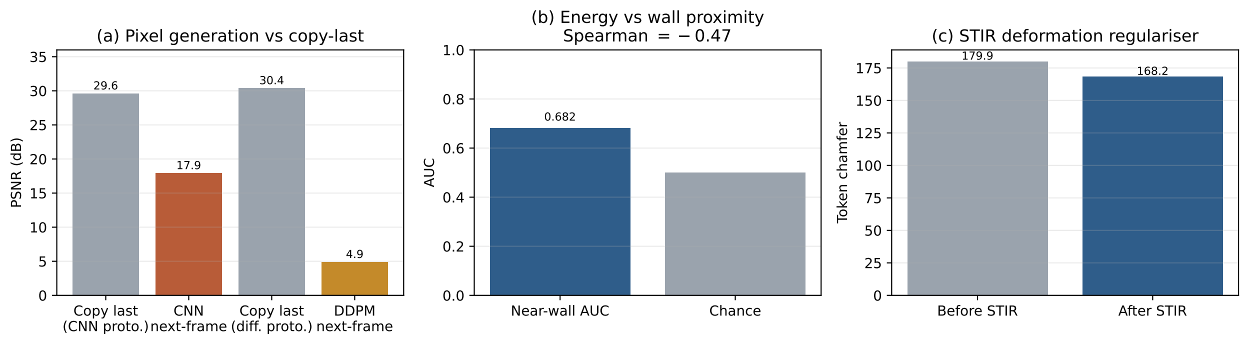

Figure 6: (a) Pixel world-model baselines lose to copy-last, consistent with Proposition 1. (b) Energy versus SCARED wall proximity. (c) STIR token chamfer before/after the regulariser (8).

seed 1), attach dataset names, and evaluate the causal-L1 checkpoint. Alignment is exact (n_{cache}=n_{rebuilt}=750). Table 6 and Figure 8 report every dataset with n≥5; SCARED and STIR appear with n=1 in the JSON and are omitted from the table.

The unified model beats persistence on every listed dataset. The largest margins are Stereo_Lap (+0.093) and ION bronchoscopy (+0.071), consistent with those videos being camera-motion-dominated and therefore the most predictable in latent space. Kvasir-Capsule has the strongest persistence (0.941) and the smallest—but still positive—margin (+0.037): capsule motion is slower, so copying the last token is already a good forecast. This is the dataset-level reading of the paper's claim: unification helps, and it does not secretly rely on one easy corpus.

4.8. Ablations

Three findings. (i) Causal autoregression is the single largest forecaster improvement (0.936 → 0.978). (ii) Residual anchoring is essential at modest scale: without it the large predictor drops below persistence on its own cache (0.768 vs 0.886). (iii) L2/L3 add little to short-horizon *forecast* (full H-JEPA 0.971 at 2k) but are what enable planning and the energy signal. Hyperparameter sensitivity on the 6k val is <0.002.

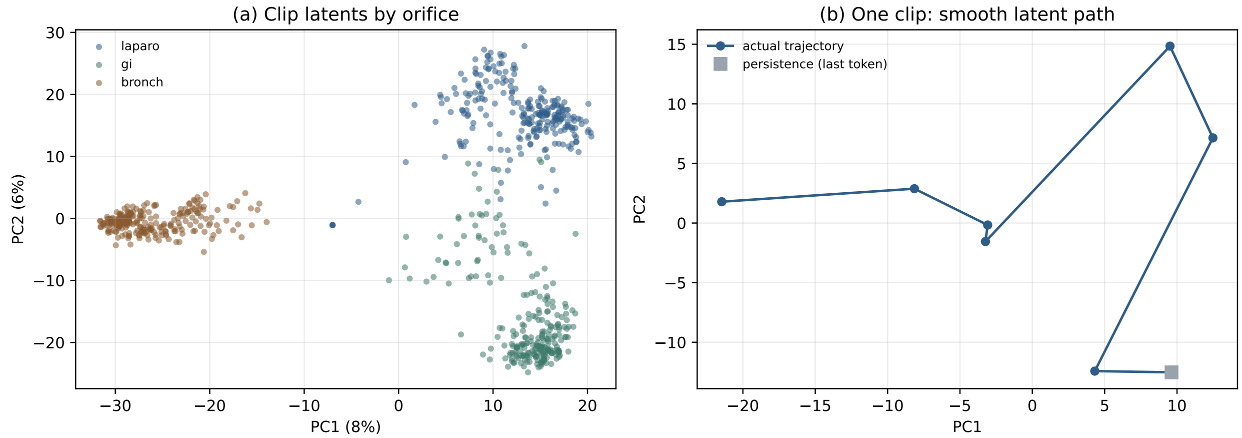

Figure 7: (a) PCA of pooled val latents coloured by orifice: a partially shared space with unsupervised domain structure. (b) One clip's latent trajectory is smooth and low-dimensional, which is why persistence is a strong short-horizon baseline.

Table 6: Per-dataset latent forecast (6,000-clip causal L1, h=4, video-level val). Source: `outputs/scale_6000_causal/per_dataset.json` (`aligned=true`). Overall 0.978 / 0.916 matches Table 3.

Dataset	Domain	n	cos	persist	Δ
ION_bronch	bronch	250	0.988	0.918	+0.071
Kvasir-Capsule	gi	206	0.978	0.941	+0.037
CholecT50	laparo	89	0.963	0.894	+0.069
Stereo_Lap	laparo	75	0.982	0.889	+0.093
endoscapes	laparo	72	0.961	0.893	+0.068
HyperKvasir	gi	44	0.970	0.924	+0.046
endovis2018_full	laparo	7	0.961	0.882	+0.079
endovis2017_full	laparo	5	0.965	0.909	+0.056
All val clips	—	750	0.978	0.916	+0.062

5. Discussion

5.1. Clinical significance

The target is anticipatory endoscopic intelligence, not another phase-recognition point. Forecast is the computational primitive behind collision warning and loss-of-view recovery. The energy head is the first quantity in this stack that correlates with a physical safety signal (camera-to-tissue distance). Unification is the data-efficient path for bronchoscopy, where labelled video is scarce. All of this remains in-silico.

5.2. What the method actually contributes

The innovations that move the numbers are, in order: residual anchoring (keeps a large predictor above persistence), causal L1 (beats GRU/Mamba), domain tokens (recover failed zero-shot transfer), and the energy head (enables MPC and tracks wall proximity). Hierarchy (L2/L3) is not a short-horizon cosine trick; it is the interface that makes planning well-posed. That is why we refuse to sell full H-JEPA as a 0.003 forecast win over L1-only.

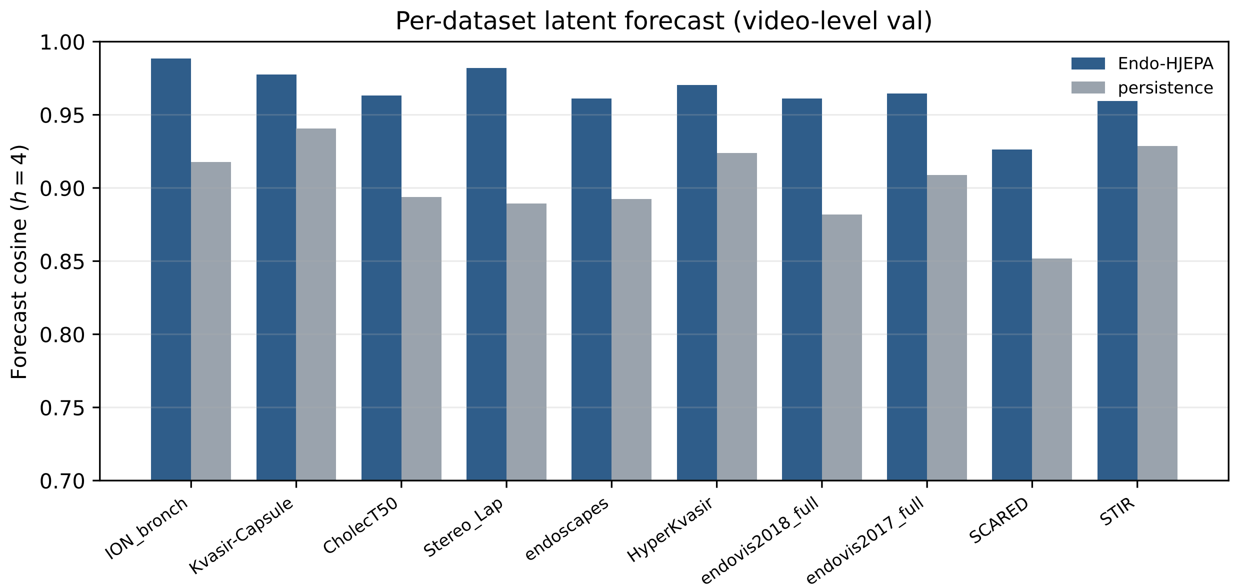

Figure 8: Per-dataset forecast corresponding to Table 6 (plus n=1 rows SCARED/STIR, not used in the table). The model beats persistence on every dataset; the largest margins are camera-motion-dominated corpora (Stereo_Lap, ION).

Table 7: Design ablations. The first five rows share the 6,000-clip protocol of Table 3. Residual on/off uses a different val cache (persist 0.886) and is not mixed into the 0.916 persist row.

Variant	cos h=4	Protocol
Persistence	0.916	6k shared cache
Query-token L1	0.936	6k
Mamba / SSM	0.971	6k
GRU	0.974	6k
Causal L1	**0.978**	6k
Full H-JEPA (L1+L2+L3+E)	0.971	2k full, forecast only
w/o residual (other cache)	0.768	persist 0.886 on that cache
with residual (same cache)	0.900	persist 0.886

5.3. Failure analysis

A small GRU is nearly competitive at h=1. Goal-directed navigation to arbitrary targets fails because latent actions encode visual change, not steer commands. Zero-shot GI/bronch transfer fails because the domains are genuinely distinct at the dynamics level. Each failure points to a concrete remedy (action supervision, domain-token adaptation) rather than to an overclaim.

5.4. Comparison to published world models

Endo-HJEPA occupies a distinct design point: hierarchical + energy-based + planning, unified across orifices, no pixel decoder. V-JEPA 2-AC, SurgRec-JEPA, EndoWAM and SVWM do not release public endoscopic-planning checkpoints, so we compare against the baselines we can run (Table 3, Table 4) and cite the others as related work, not as a fabricated leaderboard.

5.5. Limitations

(i) Training scale is far below SurgRec ($\sim 2\times10^8$ frames). (ii) Full Cholec80 and extra C3VD trajectories depend on access / Drive quota. (iii) ION has no public telemetry. (iv) Latent actions are not interpretable. (v) All planning is in-silico. (vi) Val forecast is domain-balanced, so per-dataset n is not proportional to corpus size; n=1 rows are omitted.

6. Conclusion

Endo-HJEPA brings hierarchical, energy-based joint-embedding world modelling to unified cross-orifice endoscopic video. A causal residual forecaster surpasses persistence, GRU and Mamba on a leakage-free protocol; the same stack is the only compared model that performs energy-guided latent planning; the energy head correlates with physical wall proximity; and a per-dataset decomposition shows the forecast win is not carried by a single corpus. Latent actions remain a planning device. We release the protocol, the verified metric file, and the figure generators so that every number in this paper can be regenerated from `outputs/`.

Declarations

Ethics. Public datasets are used under their original licences; ION cases are anonymised (`case_XXX`); an ethics approval number will be added before publication. All planning evaluation is in-silico. **Competing interests.** None declared. **Data/code availability.** Public datasets from their sources; the `endoworld` package, `verified_metrics.json`, and figure scripts are released with this manuscript.

Appendix A. Per-domain planning (2,000-clip full H-JEPA)

Table A.8: Forecast and planning by orifice (full H-JEPA, 2,000-clip, 250 val clips). Source: `p2000_full_causal/eval_ckpt.json`. Do not mix with Table 3 (6k causal L1).

Domain	Forecast cos (model / persist)	Plan reach (%)	n
Laparoscopy	0.960 / 0.895	95.2	84
GI endoscopy	0.970 / 0.935	98.8	83
Bronchoscopy	0.984 / 0.918	100.0	83

Appendix B. Per-class phase recognition

Appendix C. Reproducibility

Forecast: `endoworld.world.train` and `endoworld.eval.per_dataset`. Planning / energy: `endoworld.eval.eval_ckpt`, `scared_collision`, `scared_navigation`. Recognition:

Table B.9: Per-class phase accuracy (V-JEPA 2 frozen, official split, 3-seed mean).

Phase	Acc
Preparation	0.80
Calot triangle dissection	0.69
Clipping & cutting	0.67
Gallbladder dissection	0.74
Gallbladder packaging	0.67
Cleaning & coagulation	0.50
Gallbladder retraction	0.80

`endoworld.eval.cholect50_probe`. Figures: `docs/endohjepa/make_figures.py` reads only `verified_metrics.json`. Encoder caches are shared across ablations; probe heads use a fixed seed.

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